

E-commerce Shopping Behavior Analysis

Analysis of 3,900 purchases to surface spending patterns, segments, product preferences, and subscription behavior for strategic decisions.



Project Snapshot

Dataset

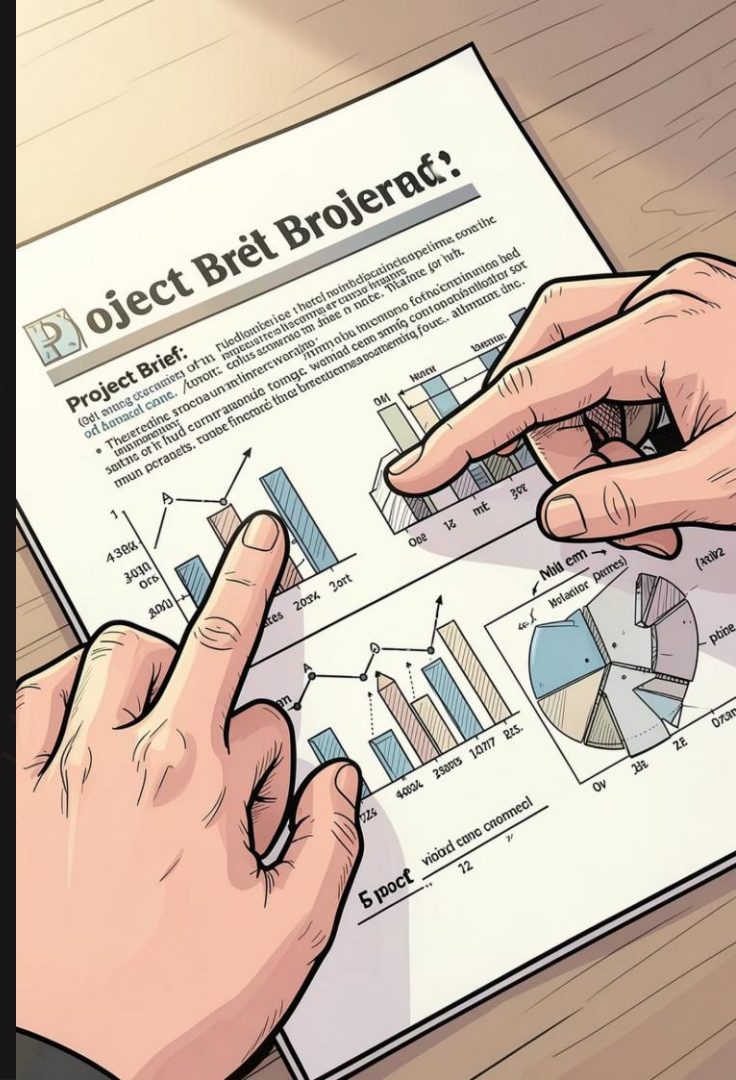
3,900 rows × 18 columns; key features include demographics, purchase details, and behavior signals.

Goal

Uncover spending patterns, segments, top products, and subscription impacts to guide strategy.

Missing Data

37 missing values in Review Rating — imputed by category median.



Data Preparation & Feature Engineering

- Loaded with pandas: inspected with `df.info()` and `describe()`.
- Renamed columns to snake_case for consistency.
- Dropped redundant `promo_code_used`; validated `discount_applied`.
- Created `age_group` bins and `purchase_frequency_days`.
- Imputed Review Rating by product-category median.



Key Descriptive Statistics



Counts & Ranges

- Customer IDs -3900 , Age mean 44.07(18-70);
- Purchase Amount mean \$59.76 (20-100).



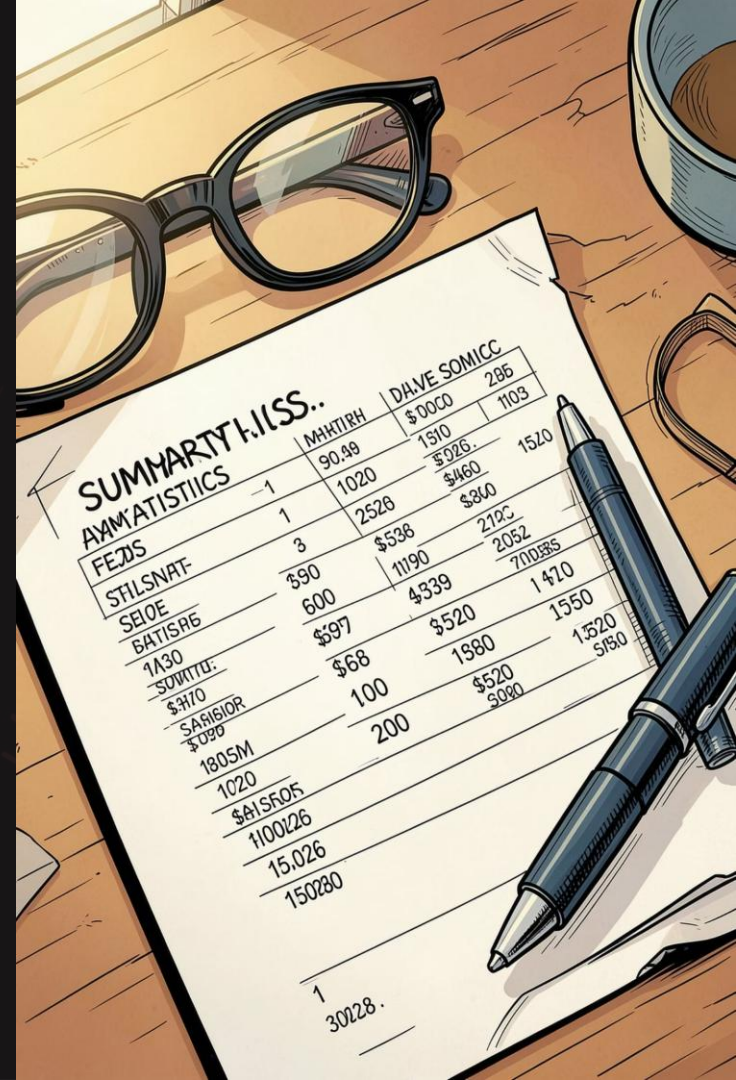
Ratings & History

Review rating mean 3.75; Previous purchases mean 3.75; 37 ratings imputed.

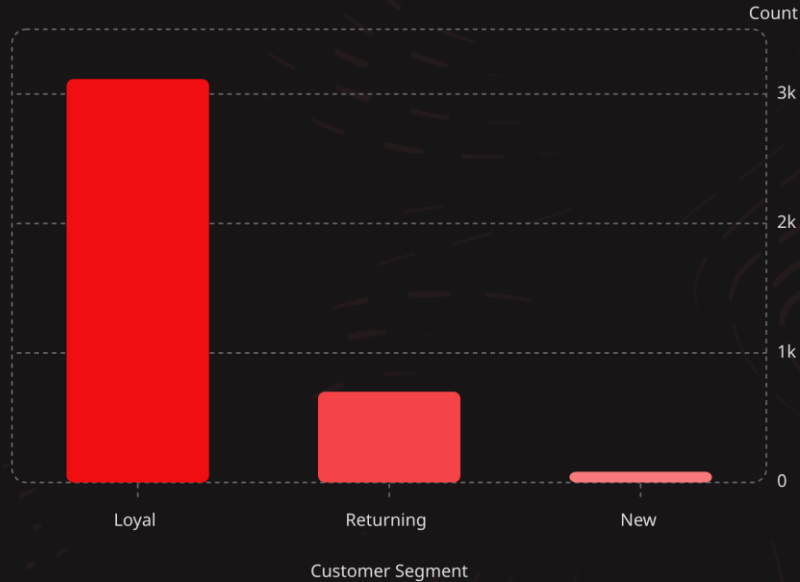


Top Values

Top item: Blouse (171), Most common location example: Montana, Top payment: PayPal.



Revenue & Segments (SQL Findings)

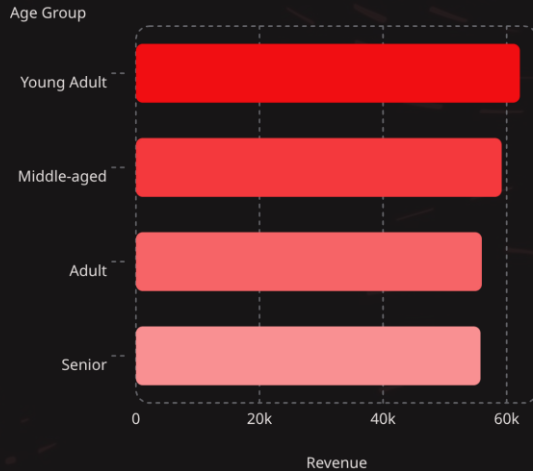


Segments: Loyal dominate (3,116). SQL shows Loyal, Returning, New counts and enables targeted strategies for retention and upsell.

- Focus loyalty programs on the large Loyal base to increase lifetime value.

Revenue by Demographics & Shipping

Shipping comparison: Express avg \$60.48 vs Standard \$58.46. Subscribers: avg spend ~\$59.49 (revenue 62,645) vs Non-subscribers avg ~\$59.87 (revenue 170,436).



Top Products & Discount Sensitivity



Top Rated Items

- Gloves (3.86) , Sandals (3.84),
- Boots (3.80) , Hat (3.78),
- Handbag (3.78)



Discount-Dependent

Highest % discounted purchases:

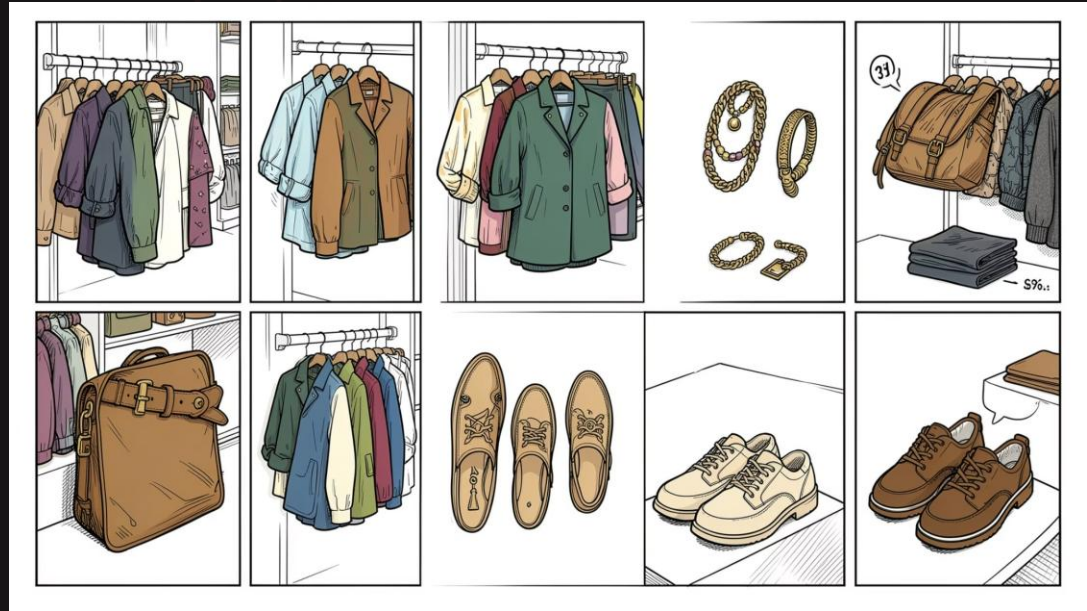
- Hat (50.00%) , Sneakers (49.66%)
Coat (49.07%) , Sweater (48.17%)
Pants (47.37%).

Top Products by Category

Accessories: Jewelry (171), Sunglasses (161), Belt (161).

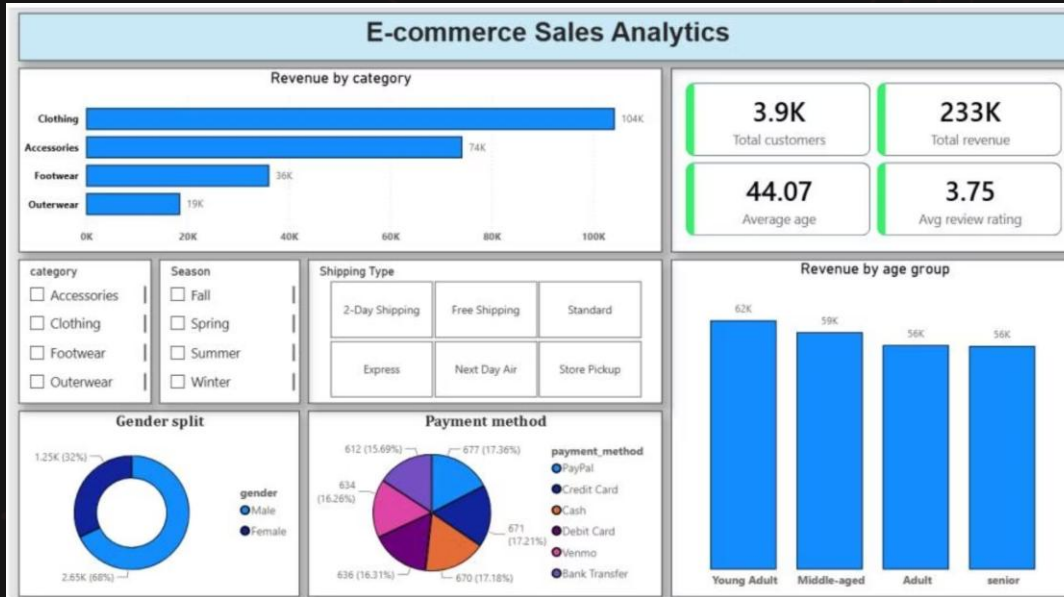
Clothing: Blouse (171), Pants (171), Shirt (169).

Footwear: Sandals (160), Shoes (150).



Promote best-sellers per category in merchandising and campaigns.

Power BI Dashboard Summary



KPIs: 3.9K total customers, \$233K total revenue, Avg age 44.07, Avg review 3.75.

Revenue by category: Clothing \$104K, Accessories \$74K, Footwear \$36K, Outerwear \$19K.

Visuals: gender donut, payment-method pie, revenue-by-age bars, filters for category/season/shipping.

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits to grow recurring revenue.

Loyalty Programs

Reward repeat buyers to convert Returning customers into Loyal segment.

Review Discount Policy

Balance promotions to drive volume without eroding margins—focus on discount-sensitive SKUs.

Product Positioning

Highlight top-rated and best-selling items in campaigns and site merchandising.

Targeted Marketing

Prioritize high-revenue age groups and express-shipping users for tailored offers.